

# Tejaswi Venumadhav Nerella

## Compact Curriculum Vitae

Associate Professor, Department of Physics, University of California, Santa Barbara

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### Appointments and Education

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**University of California, Santa Barbara**

Jul 2026–present

Associate Professor, Department of Physics

**University of California, Santa Barbara**

Jul 2020–Jul 2026

Assistant Professor, Department of Physics

**International Center for Theoretical Sciences, Bangalore**

2020–present

Visiting Professor

**Institute for Advanced Study, Princeton**

2015–2020

Member; Schmidt Fellow and John Bahcall Fellow

**California Institute of Technology**

2010–2015

Ph.D. in Physics; advisor: Christopher M. Hirata

**Indian Institute of Technology, Kanpur**

2005–2010

Integrated M.Sc. in Physics

### Research Profile

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- Theoretical astrophysics and gravitational-wave data analysis, with emphasis on compact-binary searches, parameter estimation, waveform modeling, strong lensing, neutron-star physics, and early-universe cosmology.
- Developer and user of large-scale computational methods for gravitational-wave inference and searches in non-Gaussian, non-stationary detector data.
- Publications: 52 refereed papers, 23 preprints, 3 large-collaboration papers; total h-index 32 from ADS-backed publication data.

### Computational Methods and Codes

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- Co-developer of a full gravitational-wave compact-binary search code for LIGO/Virgo data, including matched filtering, template-bank methods, ranking statistics, background estimation, and follow-up of candidates in non-Gaussian detector noise.
- Developer of *cogwheel*, a gravitational-wave parameter-estimation code for compact-binary inference, designed for efficient likelihood evaluation, sampling, and post-processing of detector data.
- Developer of additional open research codes for cosmological thermal history, sterile-neutrino dark matter, early-universe recombination physics, and related gravitational-wave/cosmology calculations.
- Experience designing large ensembles of independent analyses, validating scientific workflows across heterogeneous compute resources, and turning production-scale runs into reproducible publications and public data products.

### Selected Funding, Honors, and Awards

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- NSF, *WoU-MMA: Targeted Search for Binary Mergers with Multiple Harmonics in Gravitational Wave Data*, PI, 2023–2026.
- NSF, *WoU-MMA: Expanding the Horizons of Gravitational Wave Searches and Parameter Estimation*, PI, 2020–2024.
- Alfred P. Sloan Research Fellowship, 2023; Hellman Family Faculty Fellowship, 2023.
- John Bahcall Fellowship, Institute for Advanced Study, 2019; Schmidt Fellowship, Institute for Advanced Study, 2015–2018.
- International Fulbright Science and Technology Award, 2010; President’s Gold Medal, IIT Kanpur, 2010.

## Mentoring, Leadership, and Service

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- Research mentor to graduate students, postdoctoral scholars, undergraduate researchers, and KITP fellows in gravitational-wave astrophysics, compact-object inference, waveform modeling, and cosmology; current group projects include search pipelines, higher harmonics, eccentricity, neutron-star tides, and fast inference.
- Organizer: Prospects in Theoretical Physics 2025 program on *Gravitational Waves from Theory to Observation*; KITP 2025 program *Stellar-Mass Black Holes at the Nexus of Optical, X-ray, and Gravitational Wave Surveys*; KITP 2025 conference *The Lifecycle of Stellar Black Holes*.
- Referee for Astrophysical Journal, Astrophysical Journal Letters, Monthly Notices of the Royal Astronomical Society, Astroparticle Physics, and Physical Review D; panel reviewer for ERC, BSF, and NSF.

## Selected Publications

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1. Hegade K. R., A., Kwon, K.J., **Venumadhav, T.**, Yu, H., Yunes, N. (2026). *Relativistic and Dynamical Love Numbers*. Physical Review Letters, 136, 071401.
2. Cheung, M.H.Y., Wadekar, D., Mehta, A.K., Islam, T., Roulet, J., Berti, E., **Venumadhav, T.**, et al. (2026). *Searching for intermediate mass ratio binary black hole mergers in the third observing run of LIGO-Virgo-KAGRA*. Physical Review D, 113, 023003.
3. Mushkin, J., Roulet, J., Zackay, B., **Venumadhav, T.**, Ivashtenko, O., Wadekar, D., Mehta, A.K., Zaldarriaga, M. (2025). *Sampler-free gravitational wave inference using matrix multiplication*. Physical Review D, 112, 104025.
4. Islam, T., **Venumadhav, T.**, Mehta, A.K., Anantpurkar, I., Wadekar, D., Roulet, J., Mushkin, J., et al. (2025). *Data-driven extraction, phenomenology, and modeling of eccentric harmonics in binary black hole merger waveforms*. Physical Review D, 112, 044070.
5. **Venumadhav, T.**, Zackay, B., Roulet, J., Dai, L., Zaldarriaga, M. (2020). *New binary black hole mergers in the second observing run of Advanced LIGO and Advanced Virgo*. Physical Review D, 101, 083030.
6. Olsen, S., **Venumadhav, T.**, Mushkin, J., Roulet, J., Zackay, B., Zaldarriaga, M. (2022). *New binary black hole mergers in the LIGO-Virgo O3a data*. Physical Review D, 106, 043009.
7. Zackay, B., **Venumadhav, T.**, Dai, L., Roulet, J., Zaldarriaga, M. (2019). *Highly spinning and aligned binary black hole merger in the Advanced LIGO first observing run*. Physical Review D, 100, 023007.
8. **Venumadhav, T.**, Zackay, B., Roulet, J., Dai, L., Zaldarriaga, M. (2019). *New search pipeline for compact binary mergers: Results for binary black holes in the first observing run of Advanced LIGO*. Physical Review D, 100, 023011.
9. Zackay, B., Dai, L., **Venumadhav, T.**, Roulet, J., Zaldarriaga, M. (2021). *Detecting gravitational waves with disparate detector responses: Two new binary black hole mergers*. Physical Review D, 104, 063030.
10. Roulet, J., Chia, H.S., Olsen, S., Dai, L., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M. (2021). *Distribution of effective spins and masses of binary black holes from the LIGO and Virgo O1-O3a observing runs*. Physical Review D, 104, 083010.
11. **Venumadhav, T.**, Cyr-Racine, F.Y., Abazajian, K.N., Hirata, C.M. (2016). *Sterile neutrino dark matter: Weak interactions in the strong coupling epoch*. Physical Review D, 94, 043515.
12. Dai, L., **Venumadhav, T.**, Sigurdson, K. (2017). *Effect of lensing magnification on the apparent distribution of black hole mergers*. Physical Review D, 95, 044011.
13. Zackay, B., **Venumadhav, T.**, Roulet, J., Dai, L., Zaldarriaga, M. (2021). *Detecting gravitational waves in data with non-stationary and non-Gaussian noise*. Physical Review D, 104, 063034.
14. **Venumadhav, T.**, Dai, L., Miralda-Escudé, J. (2017). *Microlensing of Extremely Magnified Stars near Caustics of Galaxy Clusters*. The Astrophysical Journal, 850, 49.
15. Mehta, A.K., Olsen, S., Wadekar, D., Roulet, J., **Venumadhav, T.**, Mushkin, J., Zackay, B., Zaldarriaga, M. (2025). *New binary black hole mergers in the LIGO-Virgo O3b data*. Physical Review D, 111, 024049.

16. **Venumadhav, T.**, Dai, L., Kaurov, A., Zaldarriaga, M. (2018). *Heating of the intergalactic medium by the cosmic microwave background during cosmic dawn*. Physical Review D, 98, 103513.
17. Kaurov, A.A., Dai, L., **Venumadhav, T.**, Miralda-Escudé, J., Frye, B. (2019). *Highly Magnified Stars in Lensing Clusters: New Evidence in a Galaxy Lensed by MACS J0416.1-2403*. The Astrophysical Journal, 880, 58.
18. Roulet, J., **Venumadhav, T.**, Zackay, B., Dai, L., Zaldarriaga, M. (2020). *Binary black hole mergers from LIGO/Virgo O1 and O2: Population inference combining confident and marginal events*. Physical Review D, 102, 123022.
19. Dai, L., **Venumadhav, T.**, Kaurov, A.A., Miralda-Escud, J. (2018). *Probing Dark Matter Subhalos in Galaxy Clusters Using Highly Magnified Stars*. The Astrophysical Journal, 867, 24.
20. Roulet, J., Olsen, S., Mushkin, J., Islam, T., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M. (2022). *Removing degeneracy and multimodality in gravitational wave source parameters*. Physical Review D, 106, 123015.

### Selected Recent Preprints

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1. Perera, A., Zackay, B., **Venumadhav, T.** (2026). *A Multi-Mission Detection Framework for High-Energy Transients: Application to a 13-year Fermi/GBM and Swift/BAT Sample*. arXiv:2609.25672.
2. Perera, A., Zackay, B., **Venumadhav, T.** (2026). *A Coherent Search for New Galactic Magnetars using Fermi/GBM and follow-up Swift/BAT: 7 Candidates and a Large Burst Catalog*. arXiv:2609.26897.
3. Maity, K.N., Jana, S., Barsode, A., **Venumadhav, T.**, Ajith, P. (2026). *Probing Dark Matter with Strongly Lensed Binary Black Hole Mergers: Prospects in the Near Future*. arXiv:2609.05020.
4. Yu, H., Nicolini, G., Lau, S.Y., Kwon, K.J., **Venumadhav, T.**, Andersson, N., Pnigouras, P., et al. (2026). *Nonlinear hydrodynamics in spinning neutron stars: Theoretical universal relations and equilibrium solutions*. arXiv:2607.07943.